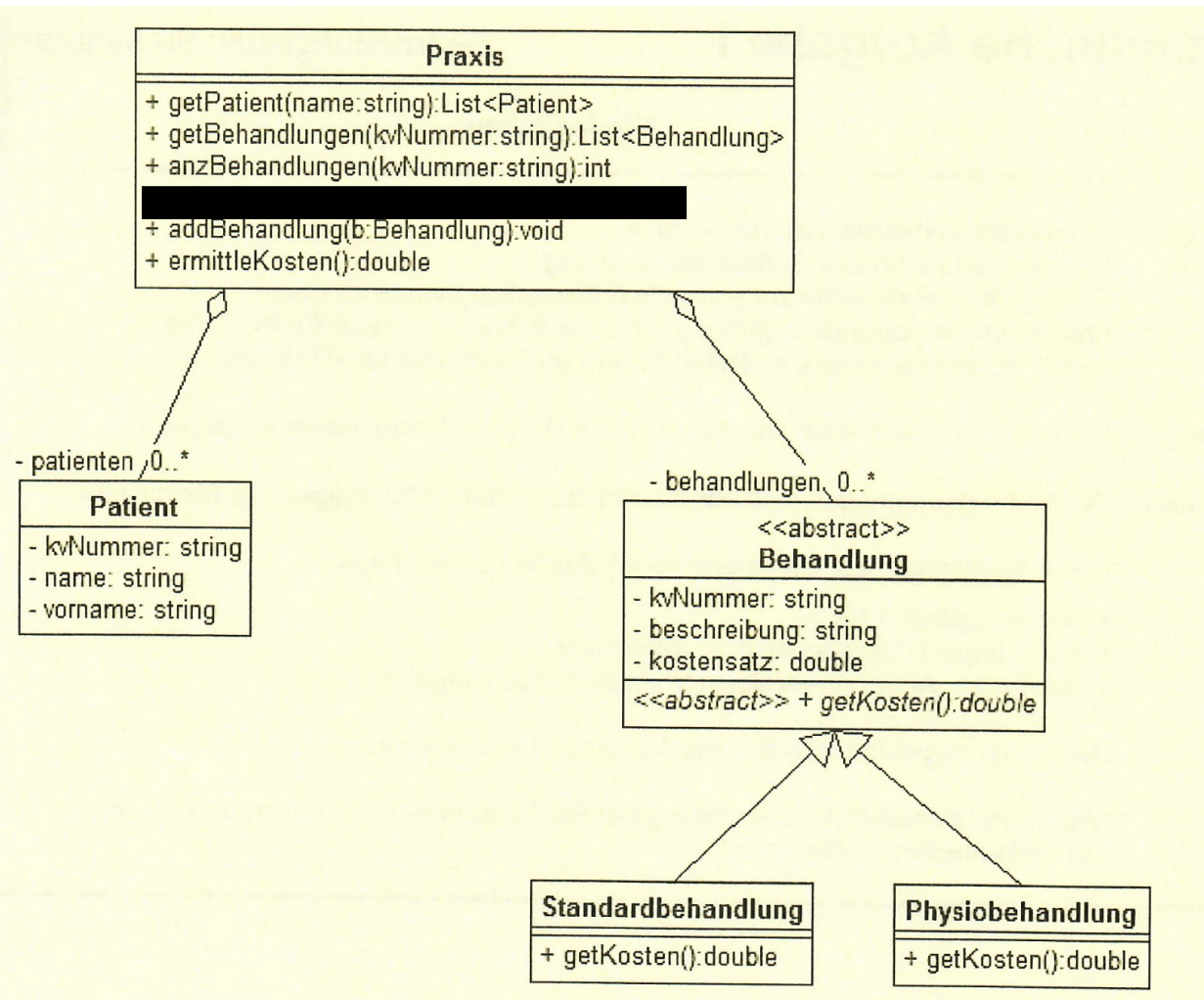


Juniorprogrammierer.de

Java: Object orientation special exercise No. 2

2024/25 – Sascha Stojanovic

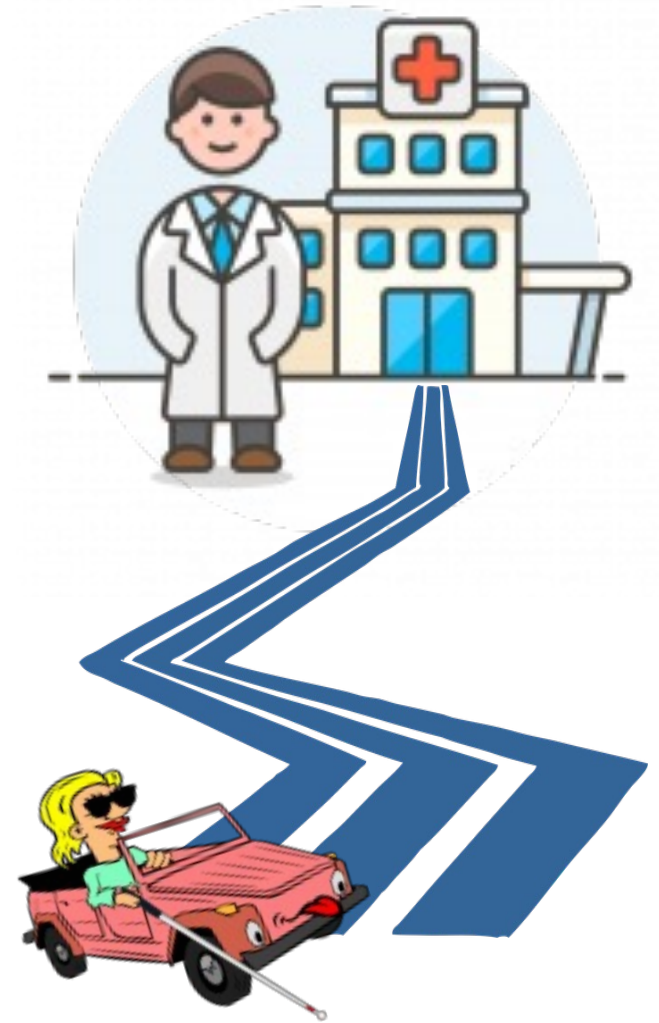
Overview



Exercise 1

Info: Every patient has a unique health insurance number (kvNummer). The assignment between treatment and patient happens through this number.

- Create the classes "Patient" and "Behandlung". Both classes should contain a parameterless constructor, a parameterized constructor, and a get() method for all attributes.



Exercise 2

- Create the classes "StandartBehandlung" and "Pyhsiobehandlung".
- For "Behandlung", create a parameterless constructor, a parameterized constructor, and the method getCosts()
- Adapt both child-classed accordingly
- The getCosts() method of the "StandardBehandlung" returns the costRate, while the getCosts() method of the PhysioTreatment returns the costRate multiplied by 1.5.



Exercise 3

Create the classe “Praxis”

- `getPatient(name: string): List<Patient>` Based on the provided name, all patients with this name (attribute "name") are identified in the patient list and returned as a list.
- `getBehandlungen(kvNummer: string): List<Behandlung>` Based on the provided health insurance number, the treatments associated with this patient are returned as a list.
- `anzBehandlungen(kvNummer: string): int` For the patient with the provided health insurance number, the number of associated treatments is determined and returned.
- `addBehandlung(b: Behandlung)` The treatment b is added to the list of treatments.
- `ermittleKosten(): double` This method calculates and returns the total costs of all treatments in the practice.



Exercise 4

Create a main-class for testing

- Add the following treatments to this object (kvNummer: A12345):
 - Physio treatment: Chiropractic; Cost rate 12.87 €
 - Physio treatment: Heat application; Cost rate 4.23 €
 - Standard treatment: Osteoarthritis; Cost rate 45.12 €
 - Standard treatment: Ultrasound; Cost rate 26.80 €
- Display the total costs of all treatments in the practice.

